

Pre-AI and post-AI design: balancing human Creativity and AI Tools in the Industrial Design Process

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Abstract

The advancement of Artificial Intelligence (AI) is reshaping industrial design by enhancing creativity and productivity. This study explores the integration of AI tools—specifically ChatGPT, DALL·E, Midjourney, and Vizcom—into the industrial design process and attempts to apply them in undergraduate industrial design education. By emphasizing the distinction between Pre-AI (divergent phase) and Post-AI (convergent phase) design stages, based on the Design Council's Double Diamond model, the impact of AI-assisted methods on creativity and the design process is examined. Through two case studies involving a Golf Cart Project and a Conceptual Car Project, we analyze the balance between human creativity and AI technology to enhance innovation while maintaining the essential role of human judgment (design decision). The findings offer preliminary insights for transforming design education and practice, highlighting the importance of foundational design skills in the era of AI.

CCS Concepts

• **Computing methodologies**; • **Artificial intelligence**; • **Knowledge representation and reasoning; Applied computing**; • **Education**; • **Interactive learning environments**;

Keywords

Industrial Design Education, AI Integration, Pre-AI Design, Post-AI Design, Human-AI Collaboration

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1 Introduction

The field of industrial design is undergoing a significant transformation due to the rapid advancement of Artificial Intelligence (AI). Designers, who traditionally relied on manual drafting and digital tools, now have access to AI tools that augment creativity and streamline the design process [1]. AI tools such as ChatGPT, DALL·E, Midjourney, and Vizcom offer new capabilities in idea generation, visualization, and iterative development. These technologies are not only changing how designers conceptualize and execute projects but also influencing the educational frameworks within which future designers are trained.

1.1 Background

AI's integration into design processes has been accelerating, with machine learning algorithms capable of analyzing vast datasets to inform design decisions. This shift parallels developments in other industries where AI is used to enhance human capabilities rather than replace them [2]. In industrial design, the synergy between human intuition and AI's computational power presents opportunities for more innovative and efficient design solutions.

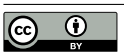
1.2 Significance of the Study

The integration of AI into industrial design presents both unprecedented opportunities and significant challenges. On one hand, AI tools can significantly enhance creativity by providing rapid ideation, visualization, and optimization capabilities, offering designers the means to explore diverse ideas quickly and efficiently. On the other hand, these benefits come with the risk of diminishing the role of human intuition and originality, particularly in the conceptual stages where personal creativity is vital.

Previous studies have primarily focused on the technical capabilities of AI in design automation and optimization. For instance, Huang and He [3] investigated the integration of Generative Adversarial Networks (GANs) into architectural design, demonstrating their potential to empower architects in creating diverse, innovative, and practical designs. Their research substantiated the feasibility of using GANs for architectural design optimization, showcasing that AI-generated architectural images exhibit significant diversity and innovation. However, they also raised concerns about the originality of AI-generated designs, as the reliance on existing data patterns could potentially limit genuine novelty, leading to a homogenization of creative outputs.

Similarly, Dorta *et al.* [4] explored the use of immersive environments like Hyve-3D in collaborative design, combining AI and virtual reality (VR) technologies to enhance the design process. Hyve-3D allows for architectural co-design within an immersive virtual space by utilizing a 3D cursor, providing users with the ability to simultaneously sketch in 3D and manipulate virtual objects intuitively using handheld tablets. This technology offers

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new possibilities for enhancing creativity and collaboration among designers. However, Dorta *et al.* emphasized that the designer's role in making critical judgments is irreplaceable; while AI and VR technologies can augment the design process, human-centered approaches remain necessary to ensure that the outcomes are not solely driven by automation but are also contextually and culturally relevant.

In the context of design education, Pei *et al.* [5] conducted an international collaborative project to understand how design education must evolve to leverage technology-driven futures, given the accelerating pace of technological change. This project compared educational provisions at a Korean design department and a UK-based design school to identify effective strategies for integrating emerging technologies. The Korean institution, situated within a specialized science and technology setting, leaned towards combining technical expertise with core design competencies, offering a holistic approach that blends theory, technology, and practice. Conversely, the UK-based institution emphasized a process-intensive curriculum, focusing on creative exploration and hands-on experience. This comparative analysis revealed that both high- and low-tech approaches coexist in these educational systems, each contributing unique strengths. Pei *et al.*'s findings underscore the importance of integrating advanced technologies into design education while maintaining a balance with foundational design skills, such as critical thinking and creative problem-solving. This balance prepares future designers to navigate both technological and human-centered challenges effectively.

Guo et al. investigated the process of co-creation between designers and generative AI, focusing on the impact of AI on designers' creative activities and the underlying mechanisms [6]. They proposed a human-AI co-creative segment model to elucidate the mechanism of AI-augmented design, emphasizing the importance of designers exerting their agency when collaborating with AI tools.

Agkathidis et al. explored the integration of AI tools like Midjourney and DALL·E as generative form-finding tools in architectural design studio teaching [7]. Their study highlighted both the advantages and challenges of utilizing AI image synthesis tools in the educational design process. While their focus was on architectural education, parallels can be drawn to industrial design education, emphasizing the need to balance rapid AI-generated designs with foundational design principles.

Sreenivasan and Suresh conducted a systematic literature review exploring the synergies between design thinking and AI [8]. Their research highlights how AI technologies (AI tools) can advance the design process by eliminating tedious tasks, improving user-centricity, and stimulating creativity. They emphasize that AI can support designers in decision-making, prototyping, and ideation processes, leading to more innovative and effective design solutions. However, they stress the importance of addressing ethical considerations such as bias in AI algorithms and data privacy to ensure responsible AI integration.

Zhou et al. focused on preparing design students to work with machine learning (ML) by building a design curriculum that enables them to understand and implement ML-enhanced products [9]. They acknowledged the challenges design students face in grasping ML's technical properties, such as its unpredictable and data-driven nature. To address this, they developed a course called Design of Information Products (DIP), which includes a teaching infrastructure and a cyclical pedagogical method tailored to design students.

Wadinambiarachchi et al. investigated the effects of generative AI on design fixation and divergent thinking in a visual ideation task [10]. Their study found that support from an AI image generator during ideation led to higher fixation on initial examples, resulting in fewer ideas with less variety and originality. This underscores potential drawbacks of AI integration, such as increased design fixation and reduced divergent thinking, for example, the limits of artificial intelligence results on human divergent thinking, highlighting the need for careful consideration in educational contexts.

Additionally, the ethical considerations surrounding the integration of AI are paramount. Mittelstadt et al. discussed bias mitigation, transparency and accountability in AI systems, emphasizing that AI algorithms trained on biased data can perpetuate or amplify existing societal biases [11]. They highlighted the need for methods to identify and mitigate bias in AI systems, to prevent discriminatory outcomes and the reinforcement of societal inequalities. Floridi and Cowls proposed a unified framework of ethical principles for AI, emphasizing the importance of transparency and accountability to build trust and ensure the responsible use of AI tools [12]. Failure to adequately address these concerns could have serious consequences, such as a lack of user trust due to the opacity of AI decision-making processes and a lack of mechanisms to hold individuals or organizations accountable for unintended consequences.

These studies collectively highlight the critical need to balance the technical advancements of AI with ethical considerations, ensuring that AI serves as an extension of human creativity without compromising originality, transparency, or accountability.

1.3 Research Gap and Contribution

In contrast to previous studies that often emphasize either the technical aspects of AI integration or the ethical considerations independently [5], [6], [9], [10], [11], this paper adopts a balanced approach. By introducing the concepts of Pre-AI and Post-AI design phases, aligned with the Design Council's Double Diamond model [13], we created a structured framework for integrating AI tools to build upon the essential role of human judgment and creativity. This study demonstrates how AI might be effectively incorporated into design education and practice, enhancing innovation while maintaining the crucial aspects of human creativity, ethical responsibility, and intuition.

Considering the distinct stages within the design process as outlined by the Double Diamond model, this approach proposes how human and AI interactions might differ across various phases of design, offering a method to harness AI's potential effectivel. As shown in 1, in the **Pre-AI Design (Divergent Phase)**, human creativity, intuition, and manual ideation are paramount. Designers tend to rely on foundational skills to establish clear objectives and initial concepts. This phase allows designers to explore a wide range of possibilities without the constraints that AI might impose.

Conversely, the **Post-AI Design (Convergent Phase)** involves incorporating AI tools to enhance the refinement and convergence

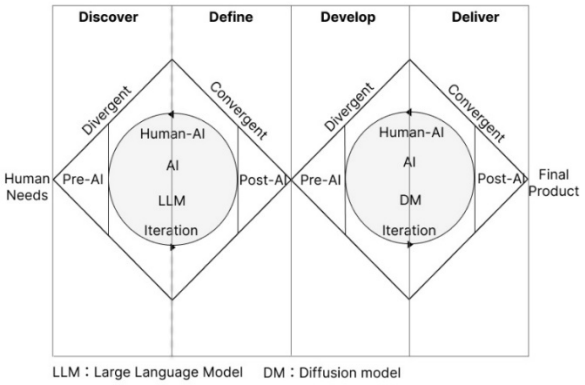


Figure 1: Pre AI design and Post AI design model, based on the double diamond model. Large language model tool: ChatGPT, diffusion model tool: DALL·E, Midjourney, Vizcom

of ideas. AI tools can process vast amounts of data and generate design options that might not be immediately apparent to human designers. Shneiderman suggests that human-centered AI can amplify human intelligence, providing tools that support creative tasks without replacing human judgment [1].

Long and Magerko have considered the potential displacement of human roles and ethical considerations such as originality and bias [22] as AI is incorporated into the design process. By exploring how AI tools can be integrated into industrial design education and practice through the distinction between Pre-AI (foundational work guiding the overall direction) and Post-AI (novel input) design phases, this study contributes practical insights into balancing human creativity with AI technology while upholding ethical standards. Understanding and integrating these phases effectively in design education can prepare future designers to harness AI's capabilities responsibly, maintaining their creative agency and ensuring ethical considerations are integral to the design process.

1.4 Objectives

The primary objective of this study is to explore when and how AI tools can be integrated into industrial design process and industrial design education to enhance creativity and efficiency without compromising the essential role of human judgment and foundational design skills.

2 Materials and Methods

2.1 Background statement

This study was conducted at Auburn University in the United States, involving undergraduate students enrolled in an industrial design course during the Fall semester of 2023 and the Spring semester of 2024. The participants comprised 11 students in Fall 2023 and 10 students in Spring 2024, including a mix of sophomores, juniors, and seniors. This diversity provided a broad spectrum of experiences with design tools and methodologies, enhancing the depth and breadth of the study. The focus was on qualitative observations and feedback from these participants.

2.2 Research Methodss

This research employed a mixed-methods approach to investigate the impact of AI tools on the design process, capturing both subjective experiences and measurable changes in design outputs. The qualitative methods primarily consisted of observational studies. Faculty members and the graduate research assistant conducted observations during studio sessions, meticulously documenting student engagement levels, the nature of interactions with both AI tools and traditional methods, and the dynamics of collaboration within groups.

Quantitative methods focused on a post-design process questionnaire using Likert scale questions designed to quantify attitudes toward the use of AI tools, perceived benefits, and challenges encountered. Open-ended questions gathered additional opinions and nuanced feedback.

The study focused primarily on qualitative observations and feedback from this group of participants (n=22). Data were collected through surveys and observations, capturing insights into students' experiences, perceptions, and the impact of AI tools on their design processes.

Ethical considerations were carefully addressed. As this study was part of regular educational activities and did not involve interventions beyond standard practice, Institutional Review Board (IRB) approval was not required. All participants were informed about the research purpose and consented to the use of their anonymized data for research purposes. Confidentiality was maintained throughout the study, and data were stored securely in accordance with university policies.

The focus on qualitative data may limit the generalizability of the findings, emphasizes the need for caution when interpreting the results, and highlights the importance of further research with larger and more diverse samples over extended periods.

3 AI in Design study

The integration of AI tools into the design process is explored through two comprehensive case studies, which illustrate how AI can influence design decisions and impact the final outcome. These studies align with the Pre-AI and Post-AI design phases of the Double Diamond Model and provide insights into the practical application of AI in design education.

Over the course of the projects, students utilized conventional drawing instruments such as pencils, markers, and sketchbooks for manual ideation, promoting tactile engagement and free-form creativity. Basic computer-aided design (CAD) software—including SolidWorks, Fusion 360, and Autodesk Alias—was employed for preliminary digital modeling to visualize concepts in three dimensions. Virtual reality (VR) tools, specifically VR headsets with Gravity Sketch, provided an immersive design experience that helped students understand spatial relationships and identify potential issues early in the process, as illustrated in Figure 2.

AI tools were integrated into the design process at specific stages. An introductory lecture covered the functionalities of each tool, tips for effective use, and ethical considerations—including discussions on originality, data privacy, and the potential biases inherent in AI systems.



Figure 2: Spring 2023 Student Work with VR headsets with Gravity Sketch for 3D modeling.



Figure 3: Spring 2023 Student Work: Golf Cart Concept Mood Board, By Midjourney (2023).

During the initial divergent and convergent phases, students used ChatGPT for brainstorming, refining ideas, and generating alternative design briefs, interacting with AI to explore different perspectives and solutions. In subsequent divergent and convergent phases, DALL·E and Midjourney were employed for text-to-image generation, assisting students in quickly visualizing concepts that would be challenging to sketch manually. Concurrently, Vizcom was used for rendering and refining sketches to enhance the visual quality of designs and facilitate rapid iteration.

In the **Pre-AI design phase**, students first reviewed the project brief and participated in brainstorming sessions that encouraged divergent thinking and collaborative ideation. They developed initial concepts through hand-drawn sketches that emphasized personal expression and creativity, and conducted research activities such as market analysis, user persona development, and investigation of design precedents to substantiate their concepts.

Upon completing the initial concepts, students transitioned to the AI design phase, integrating AI tools into their workflow. They began using ChatGPT for creative development and problem-solving, and DALL·E and Midjourney for visual exploration. Subsequently, in the **Post-AI design phase**, they combined AI-generated outputs with manual refinements to iteratively improve the design, critically evaluating the contributions of both human and AI inputs, and experimenting with CAD and VR tools for modeling.

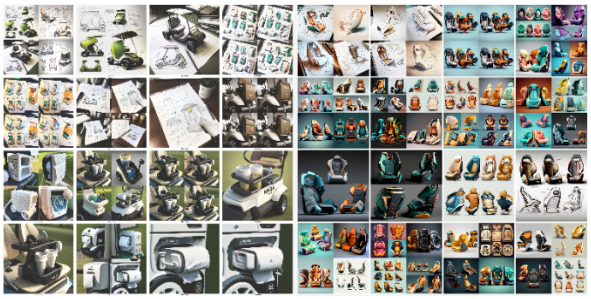


Figure 4: Spring 2023 Student Work Mood Board: Golf Cart Internal seat, luggage rack, cup holder, By DALL·E and Midjourney (2023).

3.1 Case Study 1: Golf Cart Project

In the first case study, students were tasked with designing a new golf cart for a specific brand, emphasizing functionality and aesthetics, as illustrated in 3. The project included both the **Pre-AI (Discover and Develop)** and **Post-AI (Define and Deliver)** phases, examining the impact of AI on design decisions.

During the **Pre-AI Design** Phase with the discover and develop stages of the Double Diamond model [13], students conducted extensive market research to understand and identify user needs, brand identity, and current trends in golf cart design. They gathered insights on preferences, pain points, and desired features through user interviews and surveys. Hand sketching sessions allowed them to explore forms and functions without technical constraints, promoting tactile engagement and free-form creativity.

Additionally, students explored the balance between ergonomics and aesthetics using virtual reality (VR) tools, such as VR headsets with Gravity Sketch, to visualize spatial relationships and identify potential issues early in the process. They actively investigated sustainable design applications, utilizing environmentally friendly materials and energy-saving technologies. Key design discovery made during the discover phase included focusing on user comfort with adjustable seating and luggage racks, selecting sustainable materials like recycled plastics, and designing modular components for customization.

In the **Post-AI Design** Phase, corresponding to the Define and Deliver stages, students utilized AI tools to refine their concepts. During the Define phase, they employed **ChatGPT** to explore alternative materials and solve design challenges related to sustainability. By setting up ChatGPT personas—such as having ChatGPT play the role of an engineer—they received relevant engineering suggestions. For example, when considering battery efficiency, ChatGPT recommended implementing regenerative braking systems and advanced energy storage solutions.

As shown in 4**,** during the Deliver phase, **DALL·E** and **Midjourney** were used to generate visual representations of complex concepts, such as innovative seating or unique body shapes. These AI-generated images showcased innovative shapes and features, inspiring students to incorporate new elements into their designs. Finally, students combined the AI-generated results with hand-drawn sketches to evaluate which elements to incorporate based on feasibility and consistency with the brand identity.

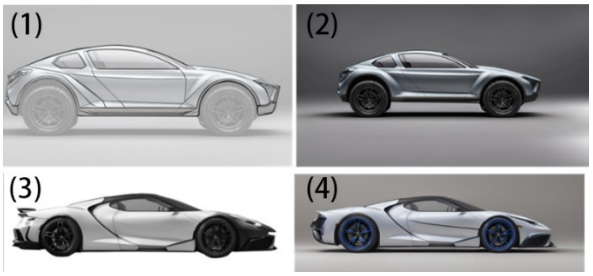


Figure 5: 2024 Spring Students work: (1) photo with underlay (2) Vizcom rendering 1 (3) Frankenstein Picture (4) Vizcom rendering 2.



Figure 6: 2024 Spring Students work: Different concepts were generated using Vizcom and stitched together using Photoshop. Finally, then used Vizcom to render the final product.

The integration of AI tools significantly enhanced the final designs. Students incorporated innovative features such as the integration of solar panels and AI-assisted navigation systems to improve functionality and sustainability. The use of AI tools facilitated rapid iteration and enabled students to visualize complex concepts more effectively. By critically evaluating the AI contributions, students significantly diversified their design proposals, ensuring that final decisions remained grounded in human judgment and thus maintained creative autonomy.

3.2 Case Study 2: Conceptual Car Project

In the second case study, students designed a car for a brand that does not currently produce vehicles, aiming to align with the brand's image and values. This project explored how AI tools could translate the brand's identity into a new product category.

During the **Pre-AI Design Phase**, students applied principles of form, function, and user experience to create mood boards and initial sketches that reflected the brand identity. They conducted a thorough brand analysis to understand the essence of the brand by examining its history, core values, and signature styles. Hand-drawn sketching sessions fostered individual expression and allowed students to explore various design directions without constraints. Key design decisions included the use of high-quality materials and elegant lines to reflect the sophistication of a luxury brand. They also identified unique selling propositions such as customization options, which would set the vehicle apart and appeal to the brand's clientele. As shown in 5, the different stages of students' conceptual construction using AI tools are demonstrated.

In the **Post-AI Design Phase**, students used **Vizcom** to enhance renderings, experiment with textures, and explore lighting effects. The AI rendering tool enabled them to achieve photorealistic visuals, improving the quality of their design presentations. They also utilized image editing software like Photoshop to blend AI-assisted imagery with their sketches, creating design drafts that incorporated diverse design elements as illustrated in 6. For example, AI-generated images introduced realistic styling elements, which the students applied to their own designs. Human judgment was crucial in selecting AI results that matched the brand identity, ensuring that the designs were authentic and cohesive.

The AI tools had a significant impact on the final vehicle designs. Students enhanced the aesthetics of their designs by improving textures and lighting effects, and adopted advanced technologies suggested by AI, such as integrated intelligent systems and sustainable materials. By critically evaluating the AI outputs, the students were able to maintain control over the design process and ensure that the final products would resonate with the target audience.

3.3 Analysis of Outcomes and Design Decisions

The integration of AI tools in both case studies led to several notable outcomes, significantly influencing the students' design processes and decisions. In the Golf Cart Project, AI tools like ChatGPT, DALL·E, and Midjourney introduced innovative concepts that expanded the students' understanding of sustainable technologies. For instance, ChatGPT, when configured to simulate an engineer's perspective, suggested implementing regenerative braking systems and advanced energy storage solutions. These suggestions prompted students to consider technologies they had not previously explored. The AI-generated images from DALL·E and Midjourney showcased unique body shapes and innovative seating layouts, inspiring students to incorporate aerodynamic designs and integrate solar panels into their concepts. By blending these AI-generated visuals with their hand-drawn sketches, students critically evaluated which elements to incorporate based on feasibility and brand alignment.

In the Concept Car Project, AI tools provided new styling elements and advanced features that aligned with the brand's identity. Using Vizcom, students achieved photorealistic renderings, enhancing the visual quality of their final products. AI-assisted imagery introduced diverse and realistic design elements, which students adapted into their own designs. By utilizing image editing software to blend AI-generated images with their sketches, they created cohesive and compelling design drafts. Human judgment was crucial in selecting AI contributions that matched the brand identity, ensuring authenticity and cohesion in the final designs.

The use of AI facilitated more informed and diverse decision-making across both projects. AI tools provided up-to-date knowledge on materials, technologies, and design trends, enabling students to make educated choices. Students learned to critically assess AI recommendations against project criteria, enhancing their analytical and decision-making skills. Instructors saw improved visualization and communication achieved through AI-powered rendering tools like Vizcom, which enhanced the aesthetic appeal

of the designs and aided in effectively conveying complex ideas. This led to more constructive feedback and collaborative refinement of concepts.

However, several challenges were encountered. Some students experienced design fixation, focusing too closely on AI outputs and potentially limiting their originality. Instructors mitigated this by encouraging exploration beyond AI suggestions and emphasizing the importance of personal creativity and divergent thinking. Ethical considerations also arose; students became aware of biases in AI-generated content and critically assessed their designs to avoid perpetuating stereotypes or cultural insensitivity. They were encouraged to evaluate the practicality of AI-generated ideas, balancing innovation with real-world constraints such as manufacturing capabilities and budget limitations. This process reinforced the importance of critical thinking and ethical responsibility in design.

3.4 Effects on Final Designs

In assessing the success of integrating of AI tools into a design project, instructors felt the final designs were more innovative and better aligned with project objectives than in projects using only traditional design methods. In the Golf Cart Project, students incorporated features (according to the principles of human function, production function and technical function) like regenerative braking, solar panels, and AI-assisted navigation systems, enhancing both functionality and sustainability that may have been missed without the AI suggestions. The visualization of complex concepts through AI tools allowed students to present their ideas more effectively, with high-quality renderings contributing to professional and compelling presentations. By critically evaluating AI contributions, students ensured that their final decisions remained grounded in human judgment, maintaining creative autonomy while embracing technological advancements.

In the Concept Car Project, the final vehicle designs exhibited a strong alignment with the brand's identity and values. Students successfully integrated advanced features and styling elements suggested by AI while ensuring consistency with the brand image. The use of premium materials, elegant lines, and innovative technologies resulted in designs that resonated with the brand language. Enhanced aesthetics achieved through AI-assisted rendering and image blending techniques elevated the overall quality of the designs. By maintaining control over the design process and selectively incorporating AI-generated ideas, students produced imaginative and feasible designs that effectively bridged the gap between the brand's traditional design language and the new automotive venture.

3.5 Reflection on Human-AI Collaboration

The case studies highlight the potential of AI tools to enhance the design process while underscoring the indispensable role of human judgment and creativity. Students recognized that AI served as a valuable resource for inspiration and problem-solving but was not a substitute for their own ideas and expertise. The collaboration between human designers and AI tools led to a synergistic relationship where each complemented the other, resulting in more refined and innovative outcomes.

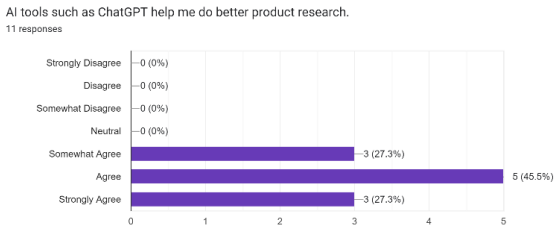


Figure 7: . A concluding horizontal bar chart for 'AI tools such as ChatGPT help me to do better product research.'

The first key reflection is the augmentation of creativity. AI tools expanded the students' creative horizons by introducing unconventional ideas and solutions, stimulating divergent thinking. Successful integration of AI required students to critically engage with the outputs, assessing the relevance, feasibility, and ethical implications of AI-generated suggestions. This critical engagement ensured that designs met project requirements and personal standards while leveraging the strengths of AI.

The importance of preserving foundational design skills was evident throughout the projects. While AI tools enhanced certain aspects of the design process, manual sketching, modeling, and conceptual development remained crucial. Maintaining a balance between AI assistance and manual creativity prevented over-reliance on technology and promoted well-rounded skill development. The experience heightened students' awareness of ethical considerations in design, particularly regarding bias, inclusivity, and cultural sensitivity. Engaging with AI tools prompted discussions about the ethical use of technology and the designer's responsibility to society.

Familiarity with AI tools prepared students for the evolving landscape of industrial design, where AI is increasingly prevalent. The projects equipped them with practical experience in integrating technology into their workflows, an essential competency in modern design professions. The human-AI collaboration demonstrated that when AI tools are integrated thoughtfully and responsibly, they can significantly enhance the design process without diminishing the essential role of the human designer.

Overall, the students' experiences underscore the value of embracing technological advancements while maintaining a strong foundation in creativity, critical thinking, and ethical awareness. This balanced approach is vital for the future of design education and practice, ensuring that designers are equipped to leverage AI effectively while upholding the integrity and humanity of their work.

4 Findings and Discussion

The integration of AI tools into the industrial design process revealed several significant advantages and limitations, explored through our case studies and data analysis.

One notable advantage was the enhancement of creativity and innovation among students. As shown in 7, the use of AI tools like DALL·E and Midjourney allowed students to generate a vast array of visual ideas rapidly [9], [10]. Most respondents (72.8%) agree

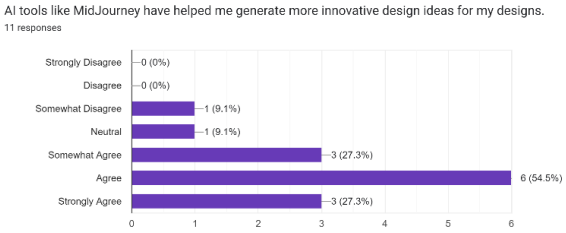


Figure 8: How student thinking about Mid Journey.

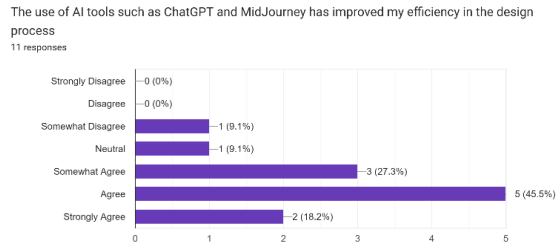


Figure 9: How AI tools like ChatGPT and MidJourney have improved their efficiency in the design process.

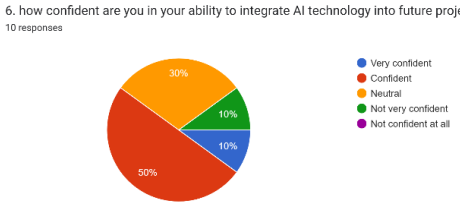


Figure 10: Student feedback on their own ability to use artificial intelligence.

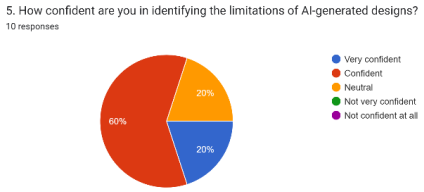


Figure 11: Student.s feedback on the limitations of AI design

or strongly agree that AI tools help to improve product research. This abundance of visual stimuli sparked new concepts and pushed students to think beyond conventional solutions. The rapid ideation facilitated by AI tools aligned well with the divergent phase of the Double Diamond model, encouraging exploration of a wide range of ideas [13], [14] as shown in Figure 6 where most respondents (81.8%) agreed or strongly agreed that AI tools like MidJourney helped them generate more innovative design ideas. Only a small fraction (9.1%) somewhat disagreed, and another 9.1% remained neutral.

As shown in 8**,** students reported that AI tools helped them overcome creative blocks by providing unexpected design suggestions. This capability was particularly beneficial in the early stages of the design process, where originality and breadth of ideas are crucial [15], [16]. Moreover, AI tools significantly improved the visualization quality of students' designs. High-quality renderings produced by tools like Vizcom enhanced the aesthetic appeal of presentations, making designs more compelling and professional [1], [17]. Improved visualizations aided in communicating complex ideas to peers and instructors more effectively, fostering better collaboration and feedback [4], [18].

Time efficiency was another advantage observed. The integration of AI tools led to accelerated design iterations compared to manual methods, allowing more time for refinement and critical evaluation [2], [19]. As shown in 9**,** Most respondents (63.7%) agreed or strongly agreed that AI tools like ChatGPT and MidJourney have improved their efficiency in the design process. A smaller portion, 9.1%, somewhat disagreed, and 9.1% remained neutral.By automating routine tasks, students could focus on higher-level design decisions, such as conceptual development and strategic aspects of their designs [8], [20]. Additionally, students developed new skills relevant to modern design practices. They gained technical proficiency in using cutting-edge AI tools, preparing them for industry demands, and demonstrated adaptive learning by integrating new technologies into their workflow [21], [22].

Despite these advantages, several limitations and challenges emerged. A significant number of students experienced a learning curve and technical barriers when using AI tools. As shown in the 10 most respondents (60%) are confident or very confident in their ability to integrate AI into future projects, while 30% are neutral, and 10% are not very confident. No one indicated that they are not confident at all. *T*herefore approximately 40% struggled with technical aspects, such as crafting effective prompts and understanding software interfaces [7], [22]. Some AI tools required high computational power or subscriptions, posing accessibility issues [1], [23].

Over-reliance on AI and potential loss of originality were also concerns. There was a risk of students becoming overly dependent on AI-generated content, leading to designs that lacked personal touch and depth [24], [25]. AI-generated designs sometimes appeared superficial, as they were based on existing data patterns rather than original thought [7], [8], [16].

Ethical and legal concerns were prominent among students. Uncertainty regarding the ownership of AI-generated content raised questions about plagiarism and copyright infringement [11], [12]. Additionally, AI models trained on biased datasets sometimes produced outputs that perpetuated stereotypes or excluded certain user groups, highlighting the need for awareness of bias in AI outputs [26], [27]. As shown in the 11, the majority of respondents (80%) are either confident or very confident in identifying the limitations of AI-generated designs, while 20% remain neutral. No respondents indicated a lack of confidence.

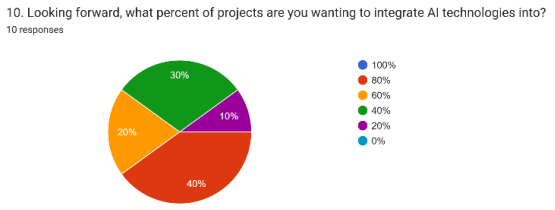


Figure 12: Feedback from students on the future integration of artificial intelligence into their projects.

Integration challenges also surfaced, such as workflow disruption when introducing AI tools mid-process, causing confusion or disjointed progression between phases [21], [19]. Balancing manual and AI methods was challenging for some students, leading to either underutilization or overdependence on AI [24], [18]. As shown in the 12, the majority of respondents (70%) are looking to integrate AI tools into at least 60% of their future projects, with 10% aiming for full (100%) integration. No one indicated that they plan to integrate AI into less than 40% of their projects.

To balance these advantages and limitations, strategic integration of AI tools was necessary. Aligning AI integration with the Double Diamond model helped introduce AI tools during the convergent phases, where refinement and selection are key [13], [14]. Instructor guidance played a crucial role in mitigating challenges, ensuring students used AI as a complement rather than a replacement [4], [21], [19]. Emphasizing foundational skills was essential in maintaining creative ownership. Continuous emphasis on manual sketching and ideation ensured that students preserved their foundational design skills [15], [18]. Encouraging critical evaluation of AI outputs helped students discern quality and relevance, preventing blind acceptance of AI-generated suggestions [1], [8], [16].

Ethical education was integral to addressing concerns. Awareness workshops on ethics and legal considerations in AI raised student awareness about intellectual property and responsible use [11], [12]. Educating students on recognizing and mitigating biases in AI outputs promoted more inclusive design solutions [26], [27]. By fostering an environment that values both technological advancement and ethical responsibility, students were better equipped to navigate the complexities of AI integration in design [2], [23].

5 Conclusions

This study demonstrates that the integration of AI tools into industrial design education offers significant advantages, such as enhanced creativity, improved visualization, time efficiency, and the development of relevant technical skills. By emphasizing the distinction between Pre-AI and Post-AI design phases, aligned with the Design Council's Double Diamond model, we have highlighted how strategic integration of AI tools can enhance the design process while preserving the essential role of human creativity and judgment.

5.1 Significance of Pre-AI Design

The Pre-AI design phase corresponds to the divergent stages of the Double Diamond model, where (discover) exploration and (develop) idea generation are paramount. These phases are crucial for fostering human creativity, intuition, and foundational design skills. Our findings emphasize that during the Pre-AI phases, manual ideation and sketching enable students to delve deeply into problem spaces without the constraints that AI might impose. For example, sometimes if students use AI-generated images first, they may limit their thinking within these frameworks. This human-centered approach allows designers to establish clear objectives and develop original concepts rooted in their cognitive abilities, as highlighted by Dorst and Cross [14]. By engaging in hands-on activities, students cultivate a personal connection to their work, which is essential for creative ownership and originality.

5.2 Impact of Post-AI Design

The Post-AI design phases align with the convergent stages of the Double Diamond model, focusing on define and deliver - refinement and selection of ideas. Incorporating AI tools during these phases enhances the design process by providing advanced capabilities for visualization, rapid iteration, and problem-solving. AI tools like ChatGPT, DALL·E, Midjourney, and Vizcom enable students to process vast amounts of information and generate design options that might not be immediately apparent. This leverages AI's computational power to introduce novel inputs, improve efficiency, and elevate the quality of design outputs without replacing human judgment, supporting Shneiderman's concept of human-centered AI amplifying human intelligence [1].

5.3 Balancing Human Creativity and AI Technology

Our study underscores the importance of balancing the Pre-AI and Post-AI design phases to ensure that AI serves as an extension of human creativity rather than a replacement. By maintaining a strong emphasis on foundational skills in the Pre-AI phase, students develop a deep understanding of design principles and retain creative ownership. The Post-AI phase builds upon this foundation, utilizing AI tools to augment and refine ideas. This balance addresses concerns about over-reliance on AI, potential loss of originality, and ethical considerations such as bias and intellectual property issues [24], [26], [27].

5.4 Recommendations for Educators

Educators are recommended to structure design curricula that clearly delineate the Pre-AI and Post-AI design phases, aligning them with established models like the Double Diamond. Providing support and training in both traditional design methods and AI tools ensures that students can navigate the entire design process effectively. Emphasizing manual sketching, critical thinking, and ethical awareness prepares students to integrate AI responsibly into their practice. Encouraging critical assessment of AI outputs helps students discern quality and relevance, preventing blind acceptance of AI-generated suggestions.

5.5 Future Research Directions

Future research should explore the long-term impacts of this phased approach to AI integration in design education. Investigating how the balance between Pre-AI and Post-AI design influences professional trajectories and creativity over time would provide valuable insights. Additionally, expanding studies to diverse educational contexts and disciplines can enhance the generalizability of the findings. Exploring team-based projects and collaborative dynamics with AI tools can further understand how human-AI interaction affects collective creativity.

6 Conclusion

In conclusion, emphasizing the significance of Pre-AI and Post-AI design phases provides a structured framework for integrating AI tools into industrial design education and practice. By harnessing the strengths of both human creativity and AI technology, designers can achieve more innovative and effective solutions. This balanced approach ensures that technological advancements enhance rather than diminish the crucial aspects of human creativity and intuition. By understanding and addressing the advantages and limitations in detail, educators can develop strategies that leverage AI's potential while fostering the development of well-rounded, critically thinking designers prepared for the evolving demands of the industry.

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Ethical Statement

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